FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 — SUB-STRAND 2.2: AREA OF POLYGONS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This Final Explanation is your opportunity to show everything you have learned across all 8 lessons about the area of polygons. Your goal is to answer the driving question fully and clearly, using the mathematics you have explored throughout this unit. Read each section prompt carefully, then write your response using precise mathematical language, worked examples, and real-world connections to the Nakuru land surveyor scenario. You must demonstrate that you can do more than recall formulas — you must show that you understand WHY each formula works, WHEN to use it, and HOW to apply it to an irregular polygon like the five-sided shamba. Use diagrams where helpful. Show all working. Write in a way that another Grade 10 student who missed these lessons could follow your explanation and understand it completely.

Section 1: The Problem With Irregular Polygons — Why Standard Formulas Are Not Enough	
Explain why the surveyor in Nakuru could not simply use the formula ½ × base × height to find the area of the five-sided shamba. In your answer: (a) describe what a perpendicular height is and why it is sometimes impossible to measure directly in the field, (b) explain what makes a polygon 'irregular,' and (c) describe the key strategy — polygon decomposition — that makes ANY irregular polygon solvable. Use a sketch of the five-sided shamba and show how it can be split into triangles.	The standard triangle area formula ½ × base × height requires you to know the perpendicular height — the straight-up-and-down distance from a base to the opposite vertex, measured at exactly 90°. On a real farm plot in Nakuru, this height might fall outside the plot entirely, or there may be a fence, crop rows, or uneven terrain blocking a direct measurement. This is why the formula alone is not enough in the field. An irregular polygon is any closed shape whose sides are not all equal and whose interior angles are not all equal — it cannot be described as a standard rectangle, square, or equilateral triangle. The surveyor's five-sided shamba (a pentagon) is irregular because no two adjacent sides are the same length and no two angles are the same. The key strategy is POLYGON DECOMPOSITION: you divide the irregular polygon into triangles by drawing diagonals from one vertex to all non-adjacent vertices. A polygon with n sides can always be split into $(n - 2)$ triangles. For the five-sided shamba, this gives $5 - 2 = 3$ triangles. Once you have three triangles, you can find the area of each triangle separately using the right formula for the measurements available, then add the three areas together. [Sketch: A five-sided irregular polygon labelled ABCDE with vertices at fence posts. Two diagonals are drawn from vertex A to vertices C and D, creating triangles ABC, ACD, and ADE. Each triangle is shaded a different colour.]

Section 2: Two Powerful Triangle Area Formulas — The Sine Rule Formula and Heron's Formula	
The surveyor has some triangles where two sides and the included angle are known, and other triangles where all three sides are known but no angle. Explain BOTH formulas that solve these situations. For each formula: (a) state the formula clearly, (b) explain what each letter/variable means, (c) explain the CONDITIONS under which you use it (what measurements must be available), and (d) show a fully worked numerical example using measurements that could realistically come from a farm plot near Nakuru. Use correct units throughout.	FORMULA 1 — The Sine Area Formula (use when you know two sides and the included angle, abbreviated SAS): Area = ½ × a × b × sin(C) Where: a and b are the lengths of the two known sides, and C is the angle between (included by) those two sides. The 'sin' function connects the angle to the perpendicular height without you needing to measure the height directly — sin(C) effectively calculates the height as a fraction of side b. Condition: You must know exactly TWO SIDES and the ANGLE BETWEEN THEM. Worked Example: Triangle ABC from the Nakuru shamba. The surveyor measures side AB = 45 m, side AC = 38 m, and the angle at A between them = 64°. Area = ½ × 45 × 38 × sin(64°) Area = ½ × 45 × 38 × 0.8988 Area = ½ × 1534.836 Area ≈ 767.4 m² --- FORMULA 2 — Heron's Formula (use when you know all three sides but NO angles, abbreviated SSS): Step 1: Find the semi-perimeter: s = (a + b + c) ÷ 2 Step 2: Area = √[s(s − a)(s − b)(s − c)] Where a, b, and c are the three side lengths and s is half the total perimeter (the semi-perimeter). The square root brings together all four factors to produce the area without any angle measurement at all. Condition: You must know ALL THREE SIDE LENGTHS. Worked Example: Triangle ACD from the shamba. The surveyor paces out AC = 38 m, CD = 52 m, AD = 41 m. s = (38 + 52 + 41) ÷ 2 = 131 ÷ 2 = 65.5 m Area = √[65.5 × (65.5 − 38) × (65.5 − 52) × (65.5 − 41)] Area = √[65.5 × 27.5 × 13.5 × 24.5] Area = √[65.5 × 27.5 × 13.5 × 24.5] Area = √[596,048.4375] Area ≈ 771.9 m²

Section 3: Solving the Nakuru Shamba — A Full Step-by-Step Solution	
Now bring it all together. Imagine the five-sided shamba ABCDE has the following surveyor measurements: AB = 45 m, BC = 60 m, CD = 52 m, DE = 35 m, EA = 48 m. Angle ABC = 110°, angle at A between sides AB and AE = 78°. All three sides of triangle ACD are known from prior calculations. Show the COMPLETE process of calculating the total area of the shamba by: (a) decomposing the pentagon into 3 triangles, (b) identifying which formula applies to each triangle and why, (c) showing full working for each triangle's area, and (d) summing the areas to give the total area of the shamba in m² and in hectares (1 hectare = 10,000 m²).	Step 1 — Decompose the Pentagon Draw diagonals AC and AD from vertex A. This splits the shamba ABCDE into three triangles: – Triangle 1: ABC (sides AB = 45 m, BC = 60 m, included angle B = 110°) → Use SINE FORMULA (SAS) – Triangle 2: ACD (all three sides known: AC, CD = 52 m, AD) → Use HERON'S FORMULA (SSS) – Triangle 3: ADE (sides AD, DE = 35 m, EA = 48 m, included angle A = 78°) → Use SINE FORMULA (SAS) Step 2 — Calculate AC using the Cosine Rule (needed to unlock Triangle 2) $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(B)$ $AC^2 = 45^2 + 60^2 - 2(45)(60)\cos(110°)$ $AC^2 = 2025 + 3600 - 5400 \times (-0.3420)$ $AC^2 = 5625 + 1846.8 = 7471.8$ $AC \approx 86.4$ m Step 3 — Triangle 1: ABC (Sine Formula) $Area_1 = ½ \times 45 \times 60 \times \sin(110°)$ $Area_1 = ½ \times 45 \times 60 \times 0.9397$ $Area_1 = ½ \times 2537.0 \approx 1268.5$ m² Step 4 — Find AD using Cosine Rule for Triangle ADE setup, and apply Heron's to Triangle 2: ACD For Triangle ACD: AC ≈ 86.4 m, CD = 52 m, AD ≈ 41 m (surveyor measurement) $s = (86.4 + 52 + 41) \div 2 = 179.4 \div 2 = 89.7$ $Area_2 = \sqrt{[89.7 \times (89.7-86.4) \times (89.7-52) \times (89.7-41)]}$ $Area_2 = \sqrt{[89.7 \times 3.3 \times 37.7 \times 48.7]}$ $Area_2 = \sqrt{[544{,}041.5]} \approx 737.6$ m² Step 5 — Triangle 3: ADE (Sine Formula) $Area_3 = ½ \times 41 \times 35 \times \sin(78°)$ $Area_3 = ½ \times 41 \times 35 \times 0.9781$ $Area_3 = ½ \times 1403.9 \approx 701.9$ m² Step 6 — Total Area Total Area = $Area_1 + Area_2 + Area_3$ Total Area = 1268.5 + 737.6 + 701.9 Total Area ≈ 2708.0 m² In hectares: 2708.0 ÷ 10,000 ≈ 0.271 hectares The surveyor can now report to the three siblings that their inherited shamba is approximately 2708 m², or 0.27 hectares.

Section 4: Answering the Driving Question — Choosing the Right Formula for Any Situation

Directly answer the driving question: 'How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available?' Your answer must: (a) explain the decision-making process a surveyor uses to choose between the sine formula, Heron's formula, and the basic ½bh formula, (b) present a clear decision guide (you may use a flowchart, table, or bullet-point guide) that shows WHICH formula to use given WHICH measurements are available, and (c) explain why polygon decomposition is the 'master strategy' that connects all other strategies.

A land surveyor in Nakuru can calculate the exact area of ANY irregularly shaped farm plot by following two steps: (1) decompose the polygon into triangles, and (2) apply the correct area formula to each triangle based on what measurements are available. There is always a formula that works — you never need ALL possible measurements.

DECISION GUIDE — Which Triangle Area Formula Do I Use?

Measurements Available	Formula to Use
Base + perpendicular height (right angle confirmed)	Area = ½ × b × h
Two sides + included angle (SAS)	Area = ½ × a × b × sin(C) [Sine Formula]
All three sides, no angles (SSS)	Area = √[s(s−a)(s−b)(s−c)] [Heron's Formula]

Polygon decomposition is the MASTER STRATEGY because it converts any problem — no matter how many sides the plot has — into a set of triangle problems. Since we have formula solutions for every possible combination of triangle measurements, decomposition guarantees that a solution always exists. A four-sided plot becomes 2 triangles; a seven-sided plot becomes 5 triangles. The surveyor in Nakuru never needs to be defeated by an irregular shape — they simply divide and conquer.

Section 5: Real-World Reflection — Why This Mathematics Matters in Kenya

Reflect on the real-world importance of what you have learned. In your response: (a) identify TWO other real-life Kenyan situations (beyond land surveying) where calculating the area of an irregular polygon would be essential, (b) explain one potential consequence — social, legal, or economic — of getting a land area calculation WRONG in a Kenyan context, and (c) describe one thing that surprised you or that you found

(a) Two Other Real-Life Kenyan Applications:

1. Tea Estate Planning in Kericho — Tea farm managers in Kericho County need to calculate the exact area of irregularly shaped tea fields to determine how many kilograms of fertiliser to purchase and how many tea-pickers to hire per hectare. An under-estimate wastes money; an over-estimate leaves crops under-fertilised.

2. Construction of Community Water Tanks or Fishponds near Lake Victoria — Engineers designing an irrigation channel or a fishpond on an irregular piece of land along the shores of Lake Victoria must calculate the polygon-shaped surface area to determine the volume of water the pond can hold and the amount of lining material needed.

(b) Consequence of a Wrong Calculation:

In Kenya, land is registered under the National Land Commission and errors in boundary measurements can lead to serious legal disputes between families — especially during inheritance, as in the Nakuru shamba scenario. If the surveyor under-calculates the

challenging during this unit, and explain how you worked through it.	area, one sibling may receive a smaller legal share than they are entitled to, leading to court cases, family conflict, and financial loss. Accurate mathematics is therefore not just an academic exercise — it protects people's rights and livelihoods. (c) Personal Reflection: One thing that surprised me was how Heron's Formula produces an accurate area using ONLY side lengths — no angles required at all. At first it seemed impossible that you could know the area of a triangle without knowing any of its angles. Working through the derivation helped me understand that the three side lengths already contain enough geometric information to fully define the triangle's shape, including its height. My challenge was keeping track of which triangle I was working on during decomposition, so I learned to always label vertices clearly and use a colour-coded sketch before calculating.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding — Irregular Polygons & Decomposition	Clearly and accurately explains why standard formulas are insufficient for irregular polygons. Correctly defines perpendicular height and irregular polygon. Fully explains and illustrates polygon decomposition using (n−2) triangles with a correctly labelled diagram of the five-sided shamba.	Adequately explains the limitation of ½bh for irregular polygons and describes decomposition with mostly correct reasoning. Diagram present but may have minor labelling errors. The (n−2) relationship is stated but may not be fully explained.	Shows partial understanding; may confuse perpendicular height with slant height, or describe decomposition without explaining why it works. Diagram missing or incorrect. Little connection to the Nakuru shamba context.
Mathematical Accuracy — Sine Formula and Heron's Formula	States both formulas correctly with all variables defined. Correctly identifies the conditions (SAS vs. SSS) for each formula. Both worked examples are fully shown with correct substitution, accurate computation (sin values applied correctly), and appropriate units throughout.	Both formulas are stated correctly and conditions are mostly identified. Worked examples are shown with minor arithmetic errors that do not reflect a conceptual misunderstanding. Units are mostly correct.	One or both formulas are stated incorrectly or conditions are confused (e.g., applies Heron's to a SAS triangle). Worked examples are incomplete or contain significant computational errors. Units missing or inconsistent.
Problem Solving — Full Shamba Calculation	Correctly decomposes the pentagon into 3 triangles and accurately identifies the appropriate formula for each. All three triangle areas are computed with full working shown. Areas are correctly summed and accurately converted to hectares. Solution is clearly presented and well-organised.	Pentagon is correctly decomposed and formula choice is mostly justified. At least two of the three triangle areas are correctly calculated. Final sum is attempted and conversion to hectares is shown, though there may be one arithmetic error.	Decomposition is attempted but may produce the wrong number of triangles or assign incorrect formulas. Only one triangle area is correctly calculated, or working is not shown. Final total area and/or hectare conversion is missing or incorrect.
Communication — Decision Guide and Driving Question Answer	Directly and fully answers the driving question. Decision guide (flowchart, table, or equivalent) is clearly structured, accurate, and covers all three formula conditions (base-height, SAS, SSS). The	Driving question is answered with reasonable clarity. Decision guide is present and covers at least two of the three formula conditions. The connection between decomposition and formula	Driving question is partially answered or the response is vague. Decision guide is missing, incomplete, or contains errors. The concept of decomposition as a unifying strategy is not clearly

	role of polygon decomposition as the master strategy is explained with insight and clarity.	choice is explained but may lack depth.	communicated.
Real-World Application and Reflection	Identifies two distinct, plausible Kenyan real-life applications with specific local context (names of places, industries, or communities). Consequence of error is clearly explained with social, legal, or economic reasoning rooted in Kenyan realities. Reflection is genuine, specific, and shows metacognitive awareness of the student's own learning journey.	Two real-life applications are identified and are broadly relevant to Kenya, though one may lack specific local detail. Consequence of error is explained but reasoning may be general. Reflection mentions a challenge or surprise but may not explain how it was resolved.	Only one real-life application is identified, or both are generic (not Kenyan-specific). Consequence of error is vague or missing. Reflection is superficial, formulaic, or does not connect to the unit content.